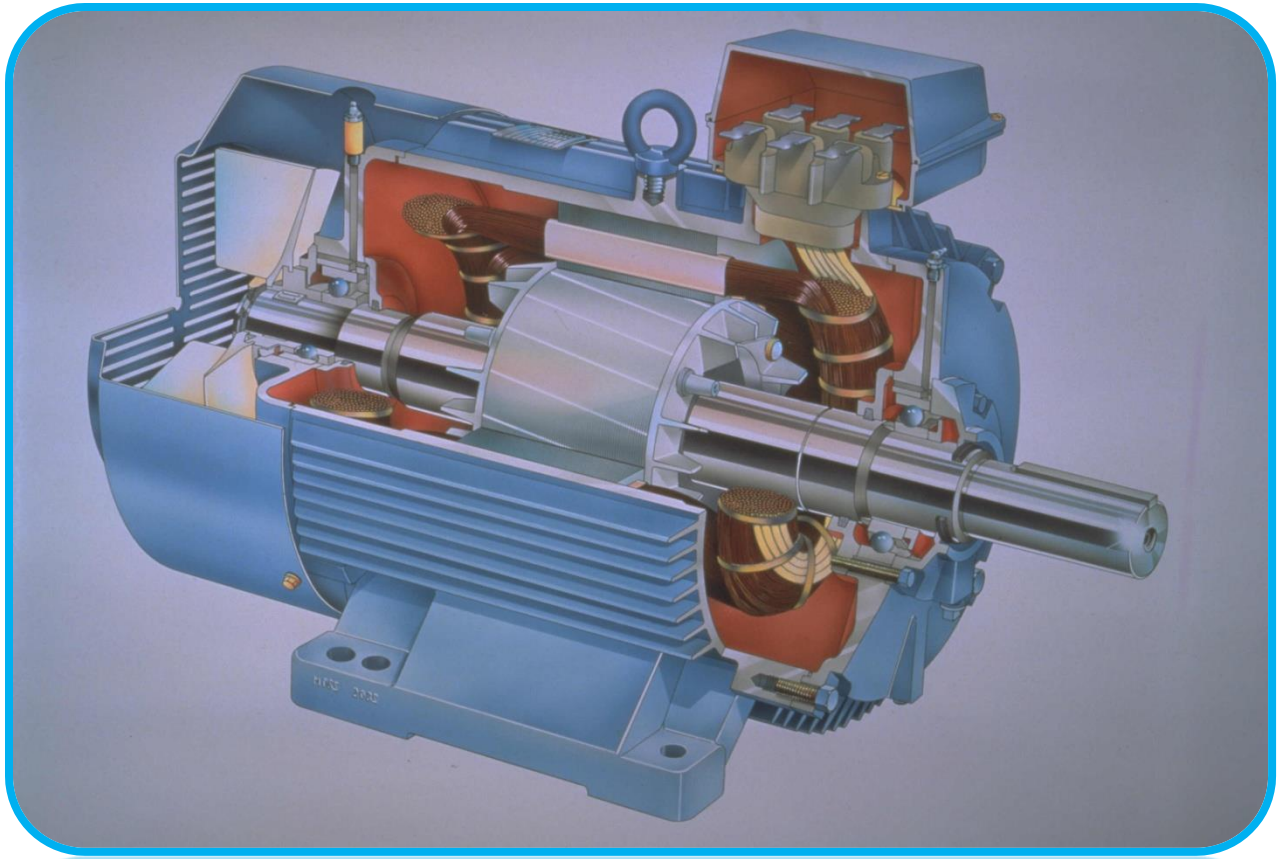
Module 4

Three phase induction Motor

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Three phase induction Motor



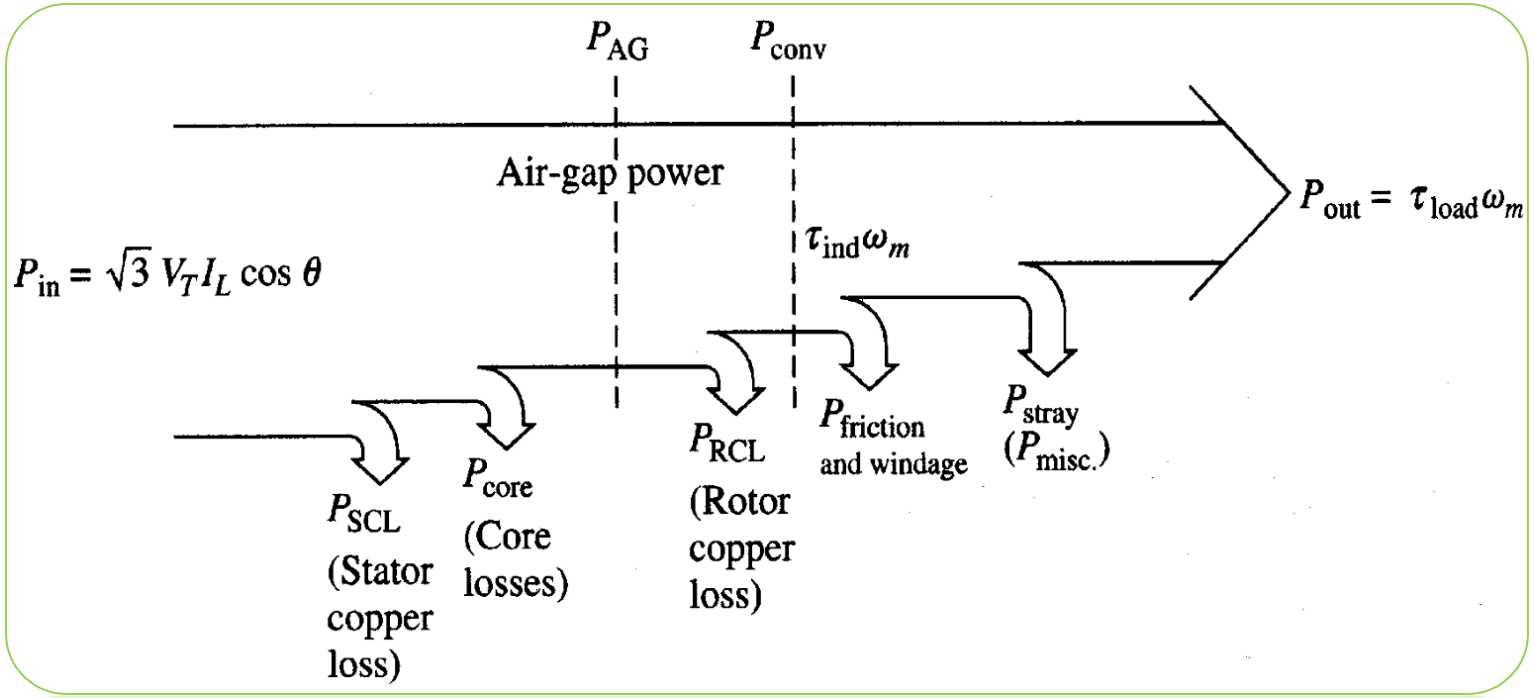
Power Flow Diagram- Losses and Efficiency

Horse power

- Another unit used to measure mechanical power is the **horse power**
- It is used to refer to the mechanical output power of the motor
- Since we, as an electrical engineers, deal with watts as a unit to measure electrical power, there is a relation between horse power and watts

$$hp = 746\ watts$$

Power flow in induction motor



Copper losses

- **Copper loss in the stator (P_{SCL}) = $I_1^2 R_1$**
- **Copper loss in the rotor (P_{RCL}) = $I_2^2 R_2$**

Core loss (P_{core})

Mechanical power loss due to friction and windage

How this power flow in the motor?

Power relations

$$P_{in} = \sqrt{3}\, V_L I_L \cos\theta = 3 V_{ph} I_{ph} \cos\theta$$

$$P_{SCL} = 3 I_1^2 R_1$$

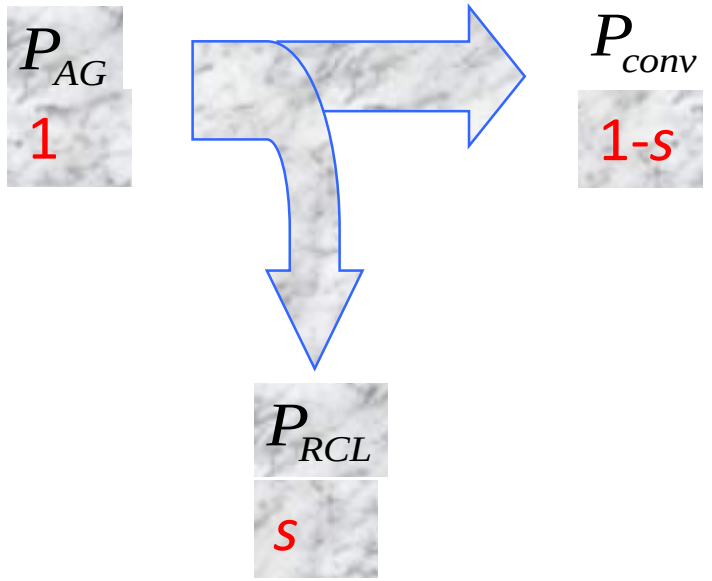
$$P_{AG} = P_{in} - (P_{SCL} + P_{core})$$

$$P_{RCL} = 3 I_2^2 R_2$$

$$P_{conv} = P_{AG} - P_{RCL}$$

$$P_{out} = P_{conv} - (P_{f+w} + P_{stray})$$

$$\tau_{ind} = \frac{P_{conv}}{\omega_m}$$



$$P_{AG} : P_{RCL} : P_{conv}$$

$$1 : s : 1\text{-}s$$

$$\therefore \quad T_g = \frac{\text{rotor input } P_2}{2\pi N_S} \qquad \ldots(2)$$

From (1) and (2), we get,

Rotor gross output $P_m = T_g \omega = T_g \times 2\pi N$

Rotor input $P_2 = T_g \omega_s = T_g \times 2\pi N_s$ ...(3)

The difference of two equals rotor Cu loss.

$\therefore$ rotor Cu loss $= P_2 - P_m = T_g \times 2\pi (N_s - N)$ ...(4)

From (3) and (4), $\dfrac{\text{rotor Cu loss}}{\text{rotor input}} = \dfrac{N_s - N}{N_s} = s$

$\therefore$ rotor Cu loss $= s \times$ rotor input $= s \times$ power across air-gap $= s P_2$ ...(5)

Also, rotor input $=$ rotor Cu loss/s

Rotor gross output, $P_m =$ input $P_2 -$ rotor Cu loss $=$ input $- s \times$ rotor input

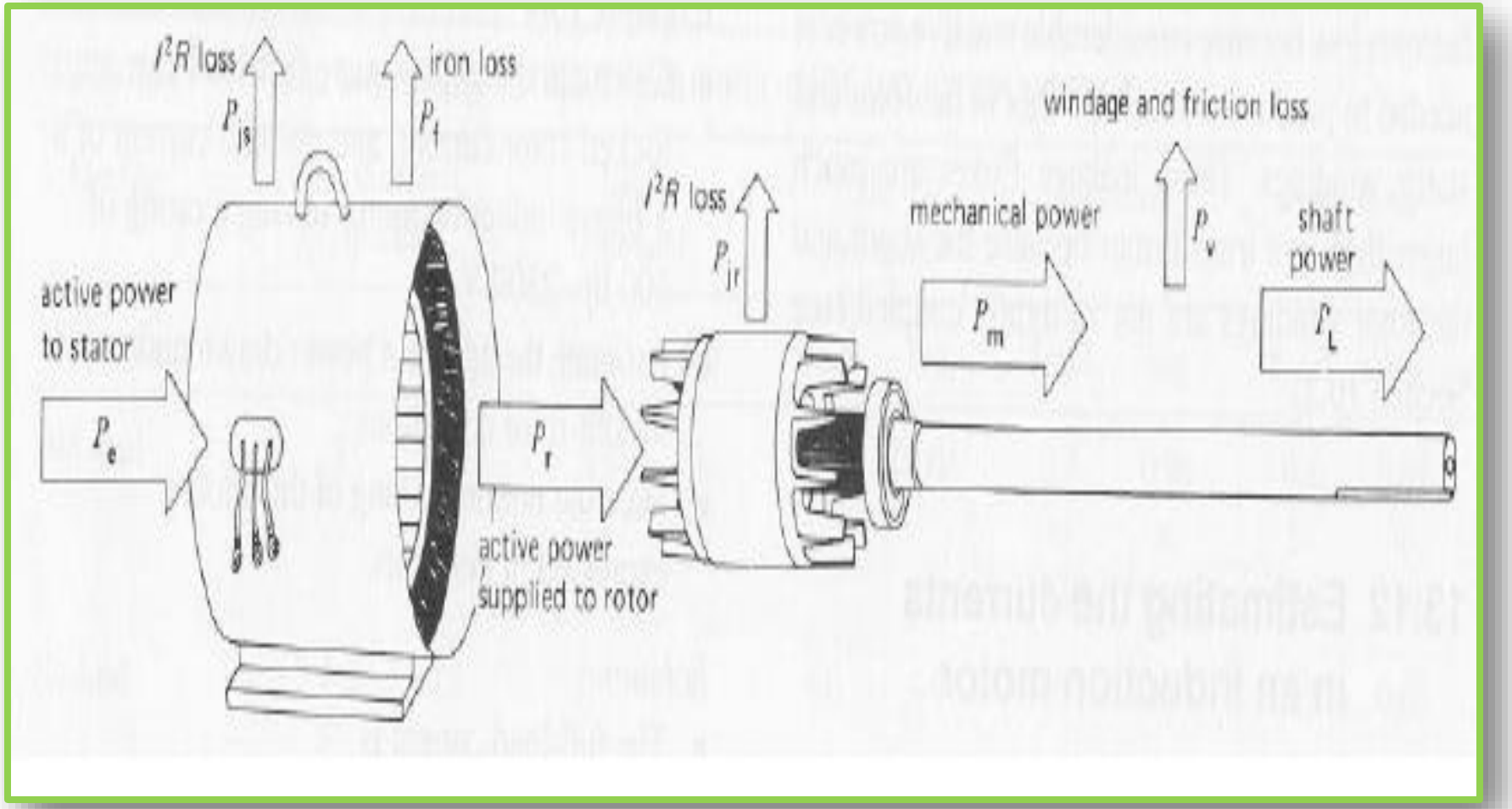
$= (1 - s)$ input P_2 ...(6)

$\therefore$ rotor gross output $P_m = (1 - s)$ rotor input P_2

or $\dfrac{\text{rotor gross output, } P_m}{\text{rotor input, } P_2} = 1 - s = \dfrac{N}{N_s}$; $\quad \dfrac{P_m}{P_2} = \dfrac{N}{N_s}$

$\therefore$ rotor efficiency $= \dfrac{N}{N_s}$ Also, $\dfrac{\text{rotor Cu loss}}{\text{rotor gross output}} = \dfrac{s}{1 - s}$

Power Flow


I^2R loss
P_{js}
iron loss
P_f
windage and friction loss
I^2R loss
P_{jr}
mechanical power
P_v
shaft power
active power to stator
P_e
P_r
P_m
P_L
active power supplied to rotor

Induction Motor Efficiency and Torque

- **The Motor efficiency:**

$$\eta = \frac{P_{out}}{P_{in}}$$

- **Motor torque:**

$$T = \frac{P_{out}}{\omega_m}$$

Example 34.35. *A 100-kW (output), 3300-V, 50-Hz, 3-phase, star-connected induction motor has a synchronous speed of 500 r.p.m. The full-load slip is 1.8% and F.L. power factor 0.85. Stator copper loss = 2440 W. Iron loss = 3500 W. Rotational losses = 1200 W. Calculate (i) the rotor copper loss (ii) the line current (iii) the full-load efficiency.* **(Elect. Machines, Nagpur Univ. 1993)**

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Solution. P_m = output + rotational loss = 100 + 1.2 = 101.2 kW

(*i*) rotor Cu loss $= \frac{s}{1-s} \times P_m = \frac{0.018}{1-0.018} \times 101.2 =$ **1.855 kW**

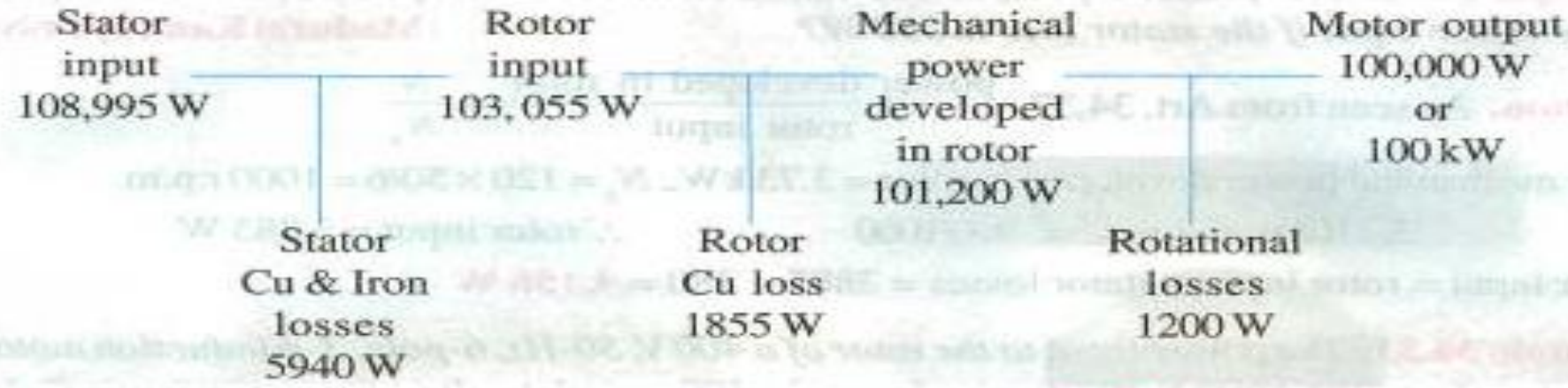
(*ii*) rotor input, P_2 = P_m + rotor Cu loss = 101.2 + 1.855 = 103.055 kW

Stator input = P_2 + stator Cu and iron losses

= 103.055 + 2.44 + 3.5 = 108.995 kW

$\therefore$ $108{,}995 = \sqrt{3} \times 3300 \times I_L \times 0.85$; $I_L =$ **22.4 A**

The entire power flow in the motor is given below.



(*iii*) F.L. efficiency = 100,000/108,995 = 0.917 or **91.7%**

Example 34.41. *An 18.65-kW, 4-pole, 50-Hz, 3-phase induction motor has friction and windage losses of 2.5 per cent of the output. The full-load slip is 4%. Compute for full load (a) the rotor Cu loss (b) the rotor input (c) the shaft torque (d) the gross electromagnetic torque.*

(Elect. Machines-II, Indore Univ. 1987)

Solution. Motor output $P_{out} = 18{,}650\ \text{W}$

Friction and windage loss $P_w = 2.5\%\ \text{of}\ 18{,}650 = 466\ \text{W}$

Rotor gross output $P_m = 18{,}650 + 466 = 19{,}116\ \text{W}$

(a) $$\frac{\text{rotor Cu loss}}{\text{rotor gross output}} = \frac{s}{1-s},\ \text{rotor Cu loss} = \frac{0.04}{(1-0.04)} \times 19{,}116 = \mathbf{796.6\ W}$$

(b) Rotor input $$P_2 = \frac{\text{rotor Cu loss}}{s} = \frac{796.5}{0.04} = \mathbf{19{,}912.5\ W}$$

(or rotor input $= 19{,}116 + 796.6 = 19{,}913\ \text{W}$)

(c) $T_{sh} = 9.55\, P_{out}/N$: $N_s = 120 \times 50/4 = 1500\ \text{r.p.m.}$

$N = (1-0.04) \times 1500 = 1440\ \text{r.p.m}$

$\therefore$ $T_{sh} = 9.55 \times 18{,}650/1440 = \mathbf{123.7\ N\text{-}m}$

(d) Gross torque $T_g = 9.55\, P_m/N = 9.55 \times 19{,}116/1440 = \mathbf{126.8\ N\text{-}m}$

(or $T_g = P_2/2\pi N_s = 19{,}913/2\pi \times 25 = 127\ \text{N-m}$)

Example 34.43. *A 6-pole, 50-Hz, 3-phase, induction motor running on full-load with 4% slip develops a torque of 149.3 N-m at its pulley rim. The friction and windage losses are 200 W and the stator Cu and iron losses equal 1,620 W. Calculate (a) output power (b) the rotor Cu loss and (c) the efficiency at full-load.* **(Elect. Technology, Mysore Univ. 1989)**

Solution. $N_s = 120 \times 50/6 = 1{,}000$ r.p.m; $N = (1 - 0.04) \times 1{,}000 = 960$ r.p.m.

Out put power $= T_{sh} \times 2\pi N = 2\pi \times (960/60) \times 149.3 =$ **15 kW**

Now, output $= 15{,}000$ W

Friction and windage losses $= 200$ W ; Rotor gross output $= 15{,}200$ W

$$\frac{P_m}{P_2} = \frac{N}{N_S} \quad \therefore \text{ rotor input } P_2 = 15{,}200 \times 1{,}000/960 = 15{,}833 \text{ W}$$

(b) $\therefore$ rotor Cu loss $= 15{,}833 - 15{,}200 =$ **633 W**

$$\left(\text{rotor Cu loss is given by : } \frac{\text{rotor Cu loss}}{\text{rotor output}} = \frac{s}{1-s}\right)$$

(c) stator output $=$ rotor input $= 15{,}833$ W

Stator Cu and iron losses $= 1{,}620$ W

$\therefore$ Stator input $P_1 = 15{,}833 + 1{,}620 = 17{,}453$ W

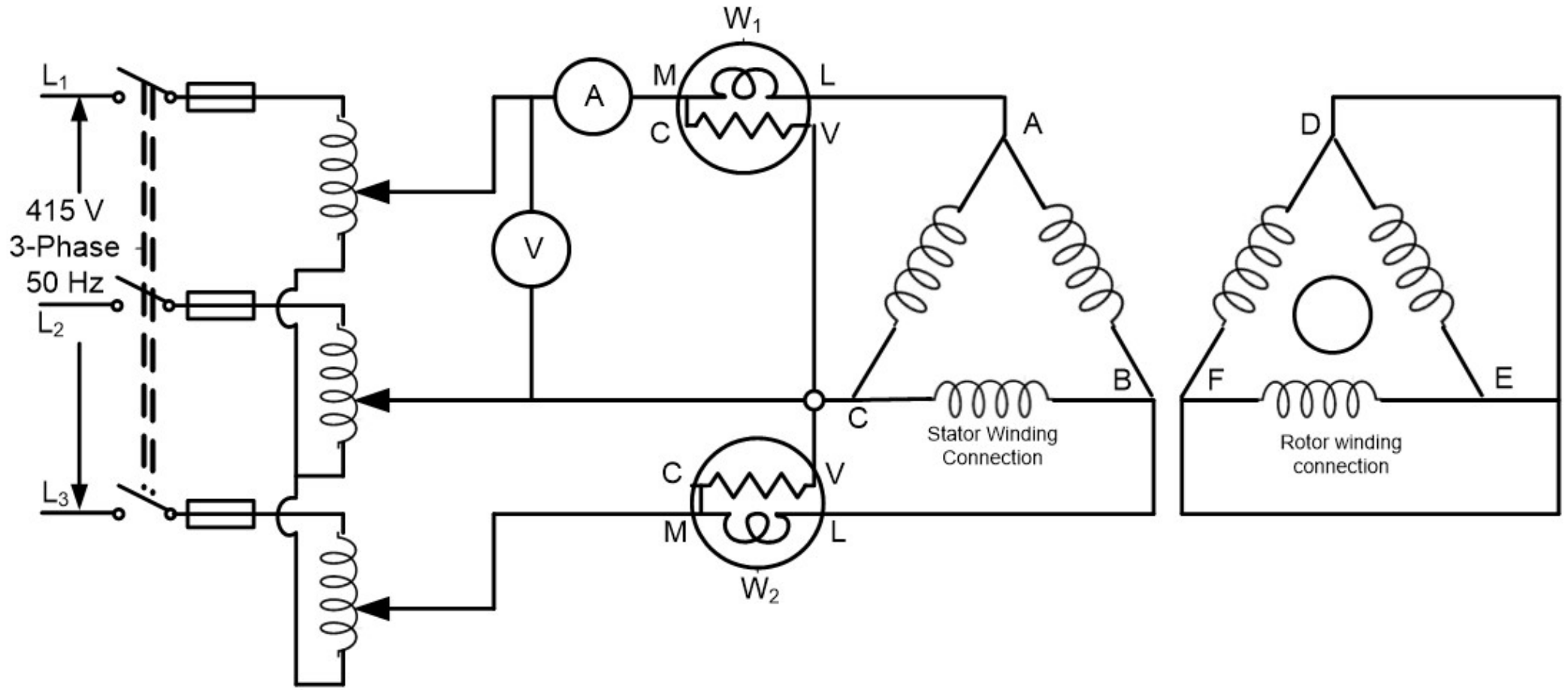
overall efficiency, $\eta = 15{,}000 \times 100/17{,}453 =$ **86%**

No Load and Blocked Rotor Test of Three phase Induction Motor

Test on Induction Motor

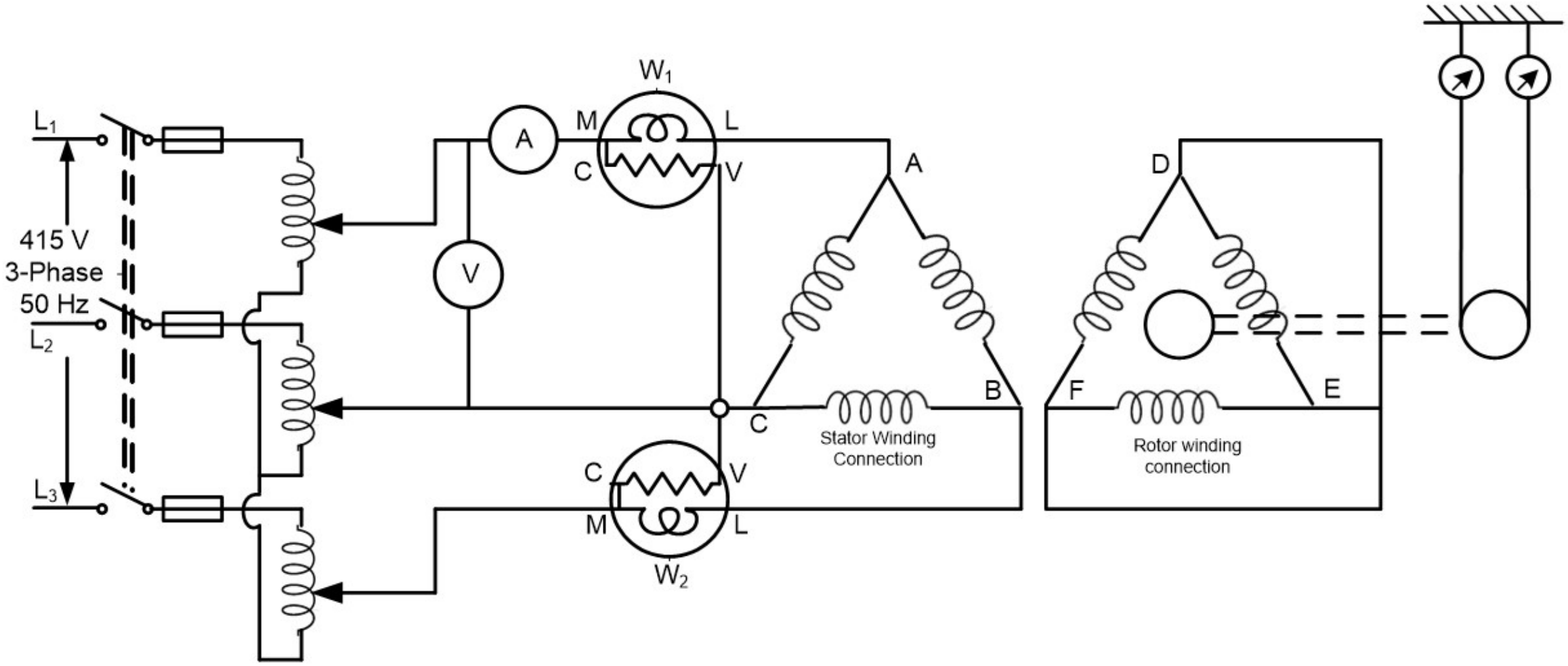
- Stator Resistance Test
- No-load Test: Stator core loss, friction & windage losses can be determined.
- Blocked Rotor Test :Stator & rotor copper losses can be determined.

No Load Test


L_1
415 V
3-Phase
50 Hz
L_2
L_3
A
V
W_1
M
L
C
V
W_2
C
V
M
L
A
B
C
Stator Winding
Connection
D
F
E
Rotor winding
connection

- No-load test is conducted to determine
  - No-load current(I_0)
  - No-load power factor($Cos\Phi_o$)
  - Windage & frictional losses(P_{wf})
  - Core loss(P_i)
  - No-load power input(P_0)
  - No-load resistance(R_0)
  - No-load reactance(X_0)

Blocked Rotor Test



- Blocked Rotor Test is conducted to determine
  - Short-circuit current with normal applied voltage to stator(I_{scN})
  - Power factor on short-circuit($Cos\Phi_{sc}$)
  - Total equivalent resistance referred to stator(R_{01})
  - Total equivalent reactance referred to stator(X_{01})

Stator Resistance:

DC resistance of stator = R_{dc}

AC resistance, $R_{ac} = 1.4 \times R_{dc}$

(a.) No Load Test:

No load voltage, (V_0) =.....V

No load current, (I_0) =.....A

No load power, $(W_0) = W_1 + W_2$ =..... watts

No load power factor, $(cos\phi_0) = \frac{W_0}{\sqrt{3}V_0 I_0}$ = ...

No load current per phase, $I_{0p} = I_0/\sqrt{3}$ =....A

Magnetizing component of current per phase, $(I_\mu) = I_{0p} sin\phi_0$ =.....A

Working component of current, $(I_w) = I_{0p} cos\phi_0$ =.....A

Therefore, $R_0 = V_0/I_w$ = ... Ω and,

$X_0 = V_0/I_\mu = \Omega$

(b.) Blocked Rotor Test:

Applied voltage, (V_{sc}) =....V

Short circuit current, (I_{sc}) =....A

Short circuit power, $(W_{sc}) = W_1 + W_2$ =..... watts

Equivalent impedance per phase referred to stator, $Z_{01} = \frac{V_{sc}}{(I_{sc}/\sqrt{3})}$ =....Ω

Equivalent resistance per phase referred to stator, $R_{01} = \frac{W_{sc}/3}{(I_{sc}/\sqrt{3})^2} = \frac{W_{sc}}{I_{sc}}$ =....Ω

The leakage reactance per phase referred to stator, $X_{01} = \sqrt{(Z_{01})^2 - (R_{01})^2}$ =....Ω

Therefore, rotor resistance referred to stator, $R_2' = R_{01} - R_{ac}$ =....Ω

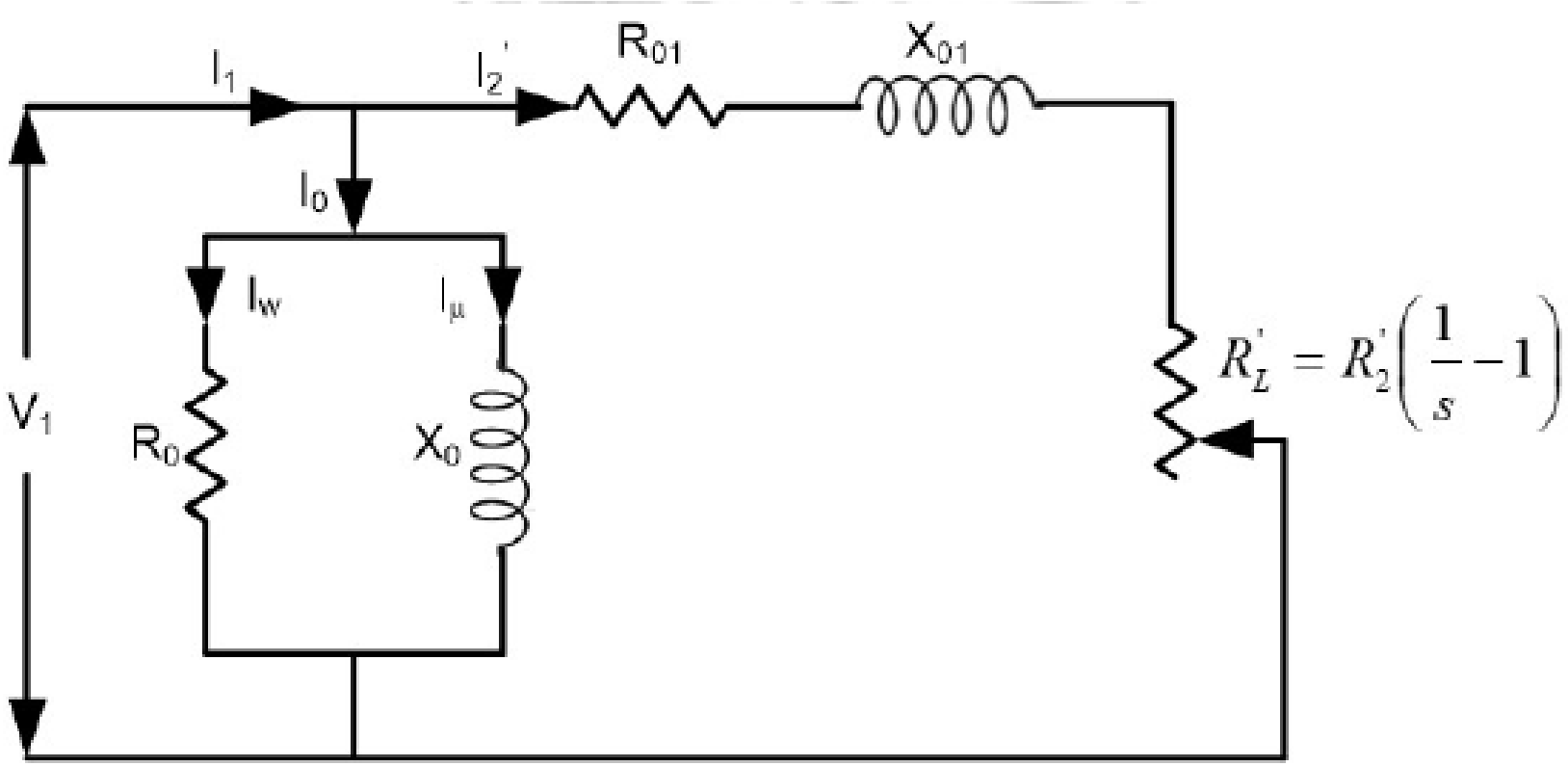


Figure 2.3: Per Phase Equivalent Circuit